Omkar Dhekane

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RESEARCH INTERESTS

Machine Learning · Information Retrieval Systems · Natural Language Processing · LLM · Computer Vision · RAG · AI agents

EDUCATION

University of Illinois Urbana Champaign, Champaign, IL — August 2024 — December 2025
Master of Computer Science (MCS) — Cumulative GPA: 4.00/4.00

Savitribai Phule Pune University, Pune, India — August 2019 — May 2023
Bachelor of Engineering in Computer Engineering — Cumulative GPA: 9.44/ 10.0
Honors: Data Science and Machine Learning
Thesis: TextGram - Towards Domain Adaptive Pre-training for LLMs

ACADEMIC EXPERIENCE

University of Illinois Urbana Champaign — Champaign, IL
Graduate Teaching Assistant — August 2025 — December 2025
CS598: Foundations of Data Curation (Fall 2025)

- Mentored 20+ students through project lifecycle (proposal to final report), providing technical feedback on code correctness and research methodology via Coursera and office hours.
- Maintained PrairieLearn auto-grader pipeline and reviewed assignments, quizzes, and exam questions to ensure accuracy and consistency
- Coordinated with instructor and co-TAs on grading rubrics and course material; monitored CampusWire to resolve student queries in real time

Coordinated Sciences Lab - Garg Group — Urbana, IL
Graduate Research & Software Assistant — May 2025 — August 2025

- Contributed to a materials science study on automated dissolution index quantification, developing an OpenCV-based image analysis pipeline to replace manual colorimetric analysis - reducing manual processing time and enabling scalable data collection for ultra-rapid material reactivity testing
- Built user interface for the prototype device, bridging experimental hardware and researcher-facing software

PUBLICATIONS

Conference Proceedings (Published)

- Magdum, V., **Dhekane, O.**, Hiwarkhedkar, S., Mittal, S., & Joshi, R. (2023). mahaNLP: A Marathi Natural Language Processing Library. *Proceedings of the 13th International Joint Conference on Natural Language Processing and the 3rd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics: System Demonstrations (**IJCNLP-AACL 2023**)*, 34–40. Association for Computational Linguistics.
- Mittal, S., Magdum, V., Hiwarkhedkar, S., **Dhekane, O.**, & Joshi, R. (2024). L3Cube-MahaNews: News-Based Short Text and Long Document Classification Datasets in Marathi. In B. R. Chakravarthi et al. (Eds.), *Speech and Language Technologies for Low-Resource Languages*, 52–63. Springer Nature.
- Hiwarkhedkar, S., Mittal, S., Magdum, V., **Dhekane, O.**, Joshi, R., Kale, G., & Ladkat, A. (2024). TextGram: Towards a Better Domain-Adaptive Pretraining. In B. R. Chakravarthi et al. (Eds.), *Speech and Language Technologies for Low-Resource Languages*, 161–173. Springer Nature.
- Gujarathi, M., Paranjape, A., Shelke, B., Gokhale, O., & **Dhekane, O.** (2023). Automatic Attendance System using a Group Image. *2023 International Conference on Emerging Smart Computing and Informatics (**ESCI 2023**)*, Pune, India. IEEE. doi:10.1109/ESCI56872.2023.10100103.
- Iqbal, M. F., Min, Y., Baten, B., Rafiuzzaman, S., **Dhekane, O.**, & Garg, N. (2026). Validation of 5-minute UR^2 on a dataset of 140+ calcined clays via low-cost colorimetry. ***Cement and Concrete Research***, 210, 108414. Elsevier. doi:10.1016/j.cemconres.2026.108414.

RELEVANT COURSES

Master's Courses

- CS441: Applied Machine Learning
- CS598: Systems in GenAI
- CS510: Advanced Information Retrieval

Bachelor's Courses

- Artificial Intelligence
- Natural Language Processing
- Information Retrieval
- Distributed Systems

OTHER EXPERIENCES

Findability.ai Mumbai, India
Data Scientist August 2023 — July 2024

- Contributed to building an ML demand forecasting system for a semiconductor HVAC client, developing a hybrid Croston-Prophet model that reduced MAPE by 10%
- Researched and experimented with novel time-series prediction models including NeuralProphet and TimeGPT to benchmark and improve forecasting accuracy

L3Cube NLP Research Lab Pune, India
Project Intern August 2022 — May 2023

- Developed **MahaNLP**, an *open-source* NLP library providing standardized preprocessing, linguistic analysis, and baseline modeling for **low-resource Indian language (Marathi)**, improving reproducibility for downstream research
- Curated and annotated Marathi long- and short-document classification datasets, enabling supervised finetuning and controlled experimentation for regional language tasks
- Conducted a comparative study of existing NLP libraries, establishing baseline benchmarks and identifying gaps in tokenization, morphology handling, and dataset coverage

Persistent Systems Pune, India
Machine Learning Intern May 2022 — August 2022

- Built a production-grade, RAG-style Extractive QA system over Legal documents, using **BM25Retriever**, ElasticSearch, **RoBERTa**-based fintuned model, and **HayStack** Framework
- Designed and deployed Flask-based REST APIs for real-time query workflows, enabling seamless integration of the QA system into internal enterprise tooling

PROJECTS

Multi-agent Research Assistant with LangGraph Champaign, IL

- Built a multi-agent research assistant using **LangGraph** StateGraph with a supervisor-critic architecture, orchestrating 4 specialized agents with conditional routing and a critic-reviser feedback loop for hallucination reduction
- Integrated **Gemini 2.5 Flash-Lite** with **LangChain** and **Tavily Search** to coordinate multi-step agentic task execution, tracking per-agent confidence scores and latency via a real-time Streamlit dashboard

Multi-label Wheat Crop disease Classification with Explainable AI Pune, India

- Finetuned a multi-label image classification pipeline for agricultural disease detection, **improving F1-score from 59.43% to 71.38%** using **DenseNet121** through targeted data augmentation strategies addressing class imbalance
- Developed a real-time, explainable inference system integrating **Grad-CAM** to support model interpretability and qualitative error analysis for field-level deployment

SKILLS

- **Programming:** Python, C++, SQL, Bash
- **Tools:** PyTorch, HuggingFace, LangChain, Ollama, CLIP, FAISS, Pinecone, OpenCV, Docker, Git
- **Soft Skills:** Research Collaboration, Teamwork, Technical Communication & Presentation, Mentorship

CERTIFICATIONS

- **Generative AI with Large Language Models**, DeepLearning.AI *Issued Jan 2024* Credential ID: R5ZNCYJM2F4W
- **Convolutional Neural Networks**, DeepLearning.AI *Issued Jun 2022* Credential ID: DJ75TLDQMFPE
- **Machine Learning**, Stanford Online *Issued Jul 2021* Credential ID: GW3TV4N9LR3A
- **Neural Networks and Deep Learning**, DeepLearning.AI *Issued Jan 2021* Credential ID: HVBM93N72JTA